Estimation:

In Estimation The goal is to make educated guesses or predictions about the characteristics of a population based on information gathered from a sample.

A **population parameter** is a numerical characteristic of an entire population, such as:

Population mean (μ)
Population variance
Population Standard Deviation (σ)

Since collecting data from every member of a population is often impractical, we use a **sample** and calculate a **statistic** (sample mean, sample proportion, etc.) to estimate the unknown population parameter.

Example

Suppose a city has 1,000,000 residents, and you want to know the average monthly income.

Population: 1,000,000 residents
Parameter: True average monthly income (μ)
Sample: 1,000 randomly selected residents
Statistic: Average income of the 1,000 sampled residents is 40,000

If the sample average income is ₹40,000, you might estimate that the population average income is approximately ₹40,000.

Types of Estimation

Point Estimation

A single value is used to estimate a population parameter.

Example:

Sample mean = ₹40,000

Estimated population mean = ₹40,000

Interval Estimation

A range of values is used to estimate the parameter, along with a **confidence level.**

Example:

Sample Mean: ₹38,500 to ₹41,500

Estimated Population Interval Mean: ₹38,500 to ₹41,500

Interval Estimation

A range of values is used to estimate the parameter, along with a confidence level.

Example:

95% confidence interval = ₹38,500 to ₹41,500

This means that, based on the sample, we are 95% confident that the true population mean lies within this range.

Sample	Sample Mean Income
Sample 1	₹40,000
Sample 2	₹39,200
Sample 3	₹40,800

Because of this variability, statisticians often report confidence intervals rather than only point estimates.

Summary

Estimating population parameters means using information from a sample to infer unknown characteristics of an entire population. The sample statistic serves as an estimate, and confidence intervals help express the uncertainty associated with that estimate.